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Screening of gall bladder cancer through infrared analysis of bile and examination of varied bile constituent composition by the disease

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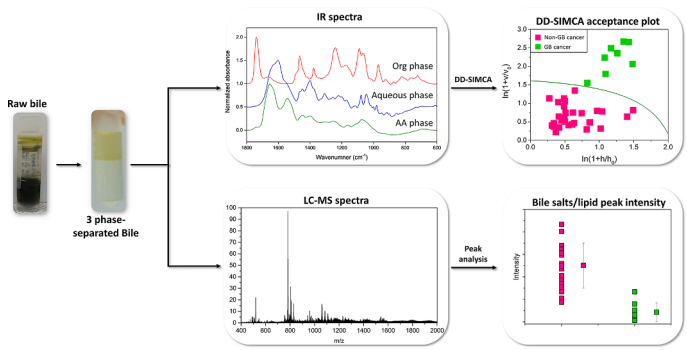
HIGHLIGHTS

- A IR-based bile analysis scheme was demonstrated for the screening of gall-bladder (GB) cancer.
- Diverse bile samples of normal/GB polyp/hepatocellular cancer/GB cancer subjects were adopted.
- Bile compositions were analyzed using LC-MS to explain the different IR feature of GB cancer.
- Bile salt/lipid levels were lower and their compositions were dissimilar in the case of GB cancer.
- IR spectrum recognized the dissimilar bile composition, so enabling the GB cancer discrimination.

GRAPHICAL ABSTRACT



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ABSTRACT

To demonstrate the infrared (IR)-based bile analysis as a reliable screening tool for gall bladder (GB) cancer, we analyzed a sample set of 37 diverse bile samples (five normal, 18 GB polyp, six hepatocellular carcinoma (HCC), and eight GB cancer subjects). Bile samples of normal subjects (control) and HCC patients were newly included to examine if IR-based bile analysis could be expanded to identify HCC. Concentrations of three bile acids and eight bile salts in the aqueous phase samples were determined in parallel and lipidomic analysis of nine lipid classes in the organic phase samples was performed using liquid chromatography-mass spectrometry. Concentrations of bile salts were lower and relative abundances of bile salts were dissimilar between GB cancer samples and remained group samples. Also, the levels of lipids such as phosphatidylcholines and phosphatidylethanolamines were again lower and their relative abundances in the organic phase of GB cancer samples were different from those of other samples. IR spectral features of the aqueous, organic, and amphiphilic aggregate phases were individually characteristic, while not descriptive enough for the thorough identification of GB cancer. Nonetheless, since they were mutually complementary to represent different metabolites in bile, the use of three phase-merged spectra was synergetic to yield the superior discrimination of GB cancer.

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1. Introduction

The feasibility of infrared (IR) spectroscopy for the discrimination of gall bladder (GB) polyp and GB cancer using bile juice as a specimen was formerly demonstrated by this research group [1]. In this previous study, raw bile was separated into an aqueous phase mostly composed of bile salts, an organic phase containing cholesterols and lipids, and a phase containing amphiphilic aggregates (AAs) comprising mostly proteins. The three-phase separation of raw bile was to acquire more component-characteristic IR spectra by decreasing compositional complexity and ultimately enhance the identification of GB cancer. Discrimination based on principal component analysis (PCA) improved when the IR spectra obtained from the three phases were merged compared to when IR spectra of each separated phase were employed individually.

However, the findings in the previous report were insufficient to support clinical applicability of the IR-based bile analysis owing to two reasons. First, due to employing the small number of bile samples (two groups of 10 GB polyp and six GB cancer samples, n = 16), the resulted discriminability of GB cancer would not be persuasive. More importantly, bile samples of normal subjects serving as controls were not included in the sample set. Second, compositional analysis of bile constituents (metabolites) enabling the explanation of dissimilar IR spectral features of the samples according to the diseases was not accomplished. It is an essential practice to support reliability for the IR spectroscopic identification of GB cancer based on multivariate discriminant analysis.

Therefore, this current study has been directed to make the IR-based bile analysis clinically reliable by addressing the mentioned two shortcomings. Initially, more diverse bile samples (five healthy, 18 GB polyp, six hepatocellular carcinoma (HCC) and eight GB cancer subjects, n = 37) were prepared as a sample set. As mentioned earlier, bile samples of healthy (normal) subjects were newly included, and bile samples from HCC patients were additionally involved to determine if IR-based bile analysis could be expanded to identify HCC, a common hepatobiliary cancer [2,3]. Next, the concentrations of three bile acids and eight bile salts in aqueous phase samples were analyzed using liquid chromatography–mass spectrometry (LC-MS). Also, untargeted lipidomics of nine different lipid classes such as phosphatidylcholines (PCs), and phosphatidylethanolamines (PEs) in the organic phase samples was performed. Then, the determined contents of bile acids, bile salts, and lipids in the normal, GB polyp, HCC, and GB cancer samples were mutually compared. Finally, data driven version of soft independent modelling of class analogies (DD-SIMCA), a one-class classification method introduced by Zontov et al. [4], was adopted to evaluate the discrimination of GB cancer samples against the samples of remained three groups using the IR spectra of aqueous, organic, and AA phase samples.

2. Materials and methods

2.1. Acquisition of bile, separation of bile, and collection of pure bile constituents

The purpose and content of our research were explained to each subject, and the study was performed under the guidelines of the Institutional Review Board of Hanyang University (HYI-18–227-3). After subject consent was obtained, bile samples were collected after cholecystectomy at Hanyang University Hospital (Seoul, Korea). Samples were immediately frozen and stored in a freezer (below −20°C) to prevent their denaturation. A total of 37 bile samples were obtained from five healthy subjects, 18 subjects with GB polyps, six patients with HCC, and eight patients with GB cancer. Bile color ranged from yellowish brown to deep green, with no noticeable fixed color differences among the four sample groups.

Each raw bile juice sample was extracted into aqueous, organic, and AA phases using a 2:1 $CHCl_3$:CH_3OH solution with multiple centrifugation steps. Details of the procedure are provided in our previous publication [1]. Hydrophilic components such as bile salts and pigments were extracted into the aqueous phase. The organic phase comprised mainly cholesterols and phospholipids, while the AA phase contained mostly proteins.

Bile salts (glycodeoxycholate (GDCA), glycochenodeoxycholate (GCDCA), glycocholate (GCA), taurodeoxycholate (TDCA), taurochenodeoxycholate (TCDCA), taurocholate (TCA)), phospholipids, and fatty acid (FA) 20:4 were obtained from Sigma-Aldrich (Avanti polar lipids). Bile acids (cholic acid (CA), chenodeoxycholic acid (CDCA), deoxycholic acid (DCA)), FA16:0, and FA18:2 were obtained from Tokyo Chemical Industry.

2.2. IR measurement of separated phases and spectral analysis

Acquisition of IR spectra of the extracted phases was accomplished according to the procedure described in our former publication [1]. Briefly, each sample was dropped onto an octadecyltrichlorosilane (OTS)-coated Si wafer and dried to remove the extraction solvent. Then, transmission IR spectra (1800–600 cm^{-1} range, resolution: 4 cm^{-1}) of the dried samples on the OTS-coated Si wafers were obtained using a Bruker LUMOS FT-IR spectrometer with a microscope (aperture size: 100 μm) and MCT detector. The baseline of each spectrum was corrected to zero at 1800 and 600 cm^{-1}, and the baseline-corrected spectrum was normalized to a maximum absorbance of 1.0 for the band with the highest absorbance. Baseline correction, normalization, and DD-SIMCA were performed using MATLAB version R2016a (The Math Works Inc., Natick, MA, USA).

2.3. Determination of bile acid/bile salt concentrations and untargeted lipidomics using LC-MS/MS

Bile acid/bile salt concentrations in the aqueous phase samples were determined using an Agilent 1290 Infinity LC system and an Agilent 6495 Triple Quadrupole MS system with an Agilent Jet Stream ESI source (Agilent Technologies, USA). Untargeted lipidomic proofing of the organic phase samples was performed on a triple TOF™ 5600 MS/MS system (AB Sciex, Canada) combined with an ACQUITY UPLC system (Waters, USA). A fully detailed description of the experimental procedure is provided in the supplementary material. Further, the relative abundances of six bile salts in each group were comparatively analyzed. Similarly, the relative peak intensity ratios of each lipid class among the four groups were investigated. To assess the statistical significance of differences in determined metabolite levels among the four groups, Mann–Whitney *U* test was conducted using Statistical Package for Social Sciences software (Version 15.0, SPSS Inc., USA).

3. Results and discussion

3.1. IR spectral features of aqueous phases and their discriminability of the diseases

Fig. 1(a) shows the average IR spectra (1800–600 cm^{-1} range) of the aqueous phase of bile samples from normal, GB polyp, HCC, and GB cancer subjects. Average spectrum of each group is arbitrarily offset for comparison. Shades superimposed on each average spectrum indicate the standard deviation in absorbance at each wavenumber. Bile salts are dominant hydrophilic components in bile and concentrations of glycine-conjugated bile salts are higher than those of taurine-conjugated bile salts [5,6]. For reference, the IR spectra of six major bile salts (GCA, GCDCA, GDCA, TCA, TCDCA, and TDCA) are presented in Fig. 2(a) and their molecular structures are shown in Fig. 3. Average ratio of glycine- and taurine-conjugated bile salt contents in bile has been reported to be approximately 3:1 under normal conditions [5], so a synthetic spectrum reflecting the 3:1 ratio was produced using the spectra of six bile salts, as shown at the bottom of Fig. 1(a). Overall shapes of the aqueous phase spectra were generally similar to that of the synthetic spectrum,

confirming the dominance of bile salts in the aqueous phase. The spectra of normal, GB polyp, and HCC samples were similar, while somewhat different from the spectrum of GB cancer, such as in the 1720–1430 cm^{-1} range (designated by the horizontal line). This observation implies that GB cancer samples have a dissimilar hydrophilic metabolite composition to normal, GB polyp, and HCC samples.

To further investigate dissimilar features among spectra, difference spectra were produced by subtracting the normal subject spectrum from the remaining spectra, as shown in Fig. 1(b). A negative difference corresponds to a greater magnitude of IR absorption compared to that in the normal subject spectrum. GB polyp and HCC samples had similar spectra, indicating that their compositions of bile salts are similar. By contrast, the shape of the GB cancer difference spectrum was dissimilar to those of the GB polyp and HCC difference spectra. More diverse negative peaks were present in the GB cancer spectrum than the other spectra, indicating that GB cancer samples have a dissimilar metabolite composition to GB polyp and HCC samples.

Fig. 4(a) through (f) show the concentrations of GCA, GCDCA, GDCA, TCA, TCDCA, and TDCA in the aqueous phase as determined by LC-MS analysis. For each bile salt, the corresponding average and standard deviation (vertical bar) are also presented to the right of the concentration data. Concentrations of these bile salts were high (millimolar range), suggesting that they made a substantial contribution to the IR spectral features. Mann-Whitney *U* test was used to assess whether pairwise differences in concentrations between groups were significant.

Concentrations of four primary bile salts (GCA, GCDCA, TCA, and

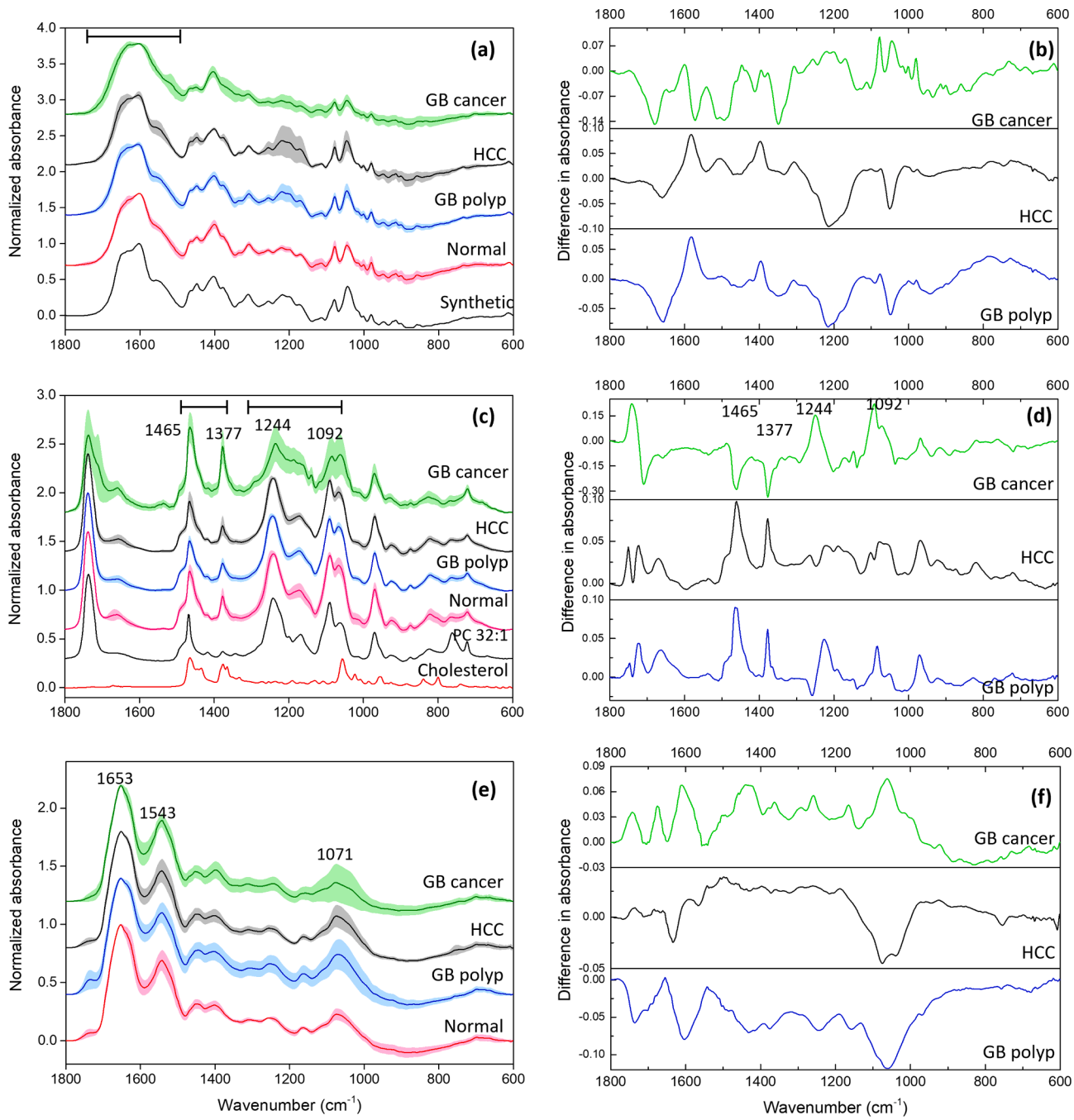

Fig. 1. Average spectra of the aqueous (a), organic (c), and AA phases (e) of normal, GB polyp, HCC, and GB cancer samples. Superimposed shades correspond to standard deviations of absorbance at each wavenumber. The difference spectra produced by subtracting the normal subject spectrum to each of the rest spectra are shown in the right column ((b), (d), and (f), respectively).

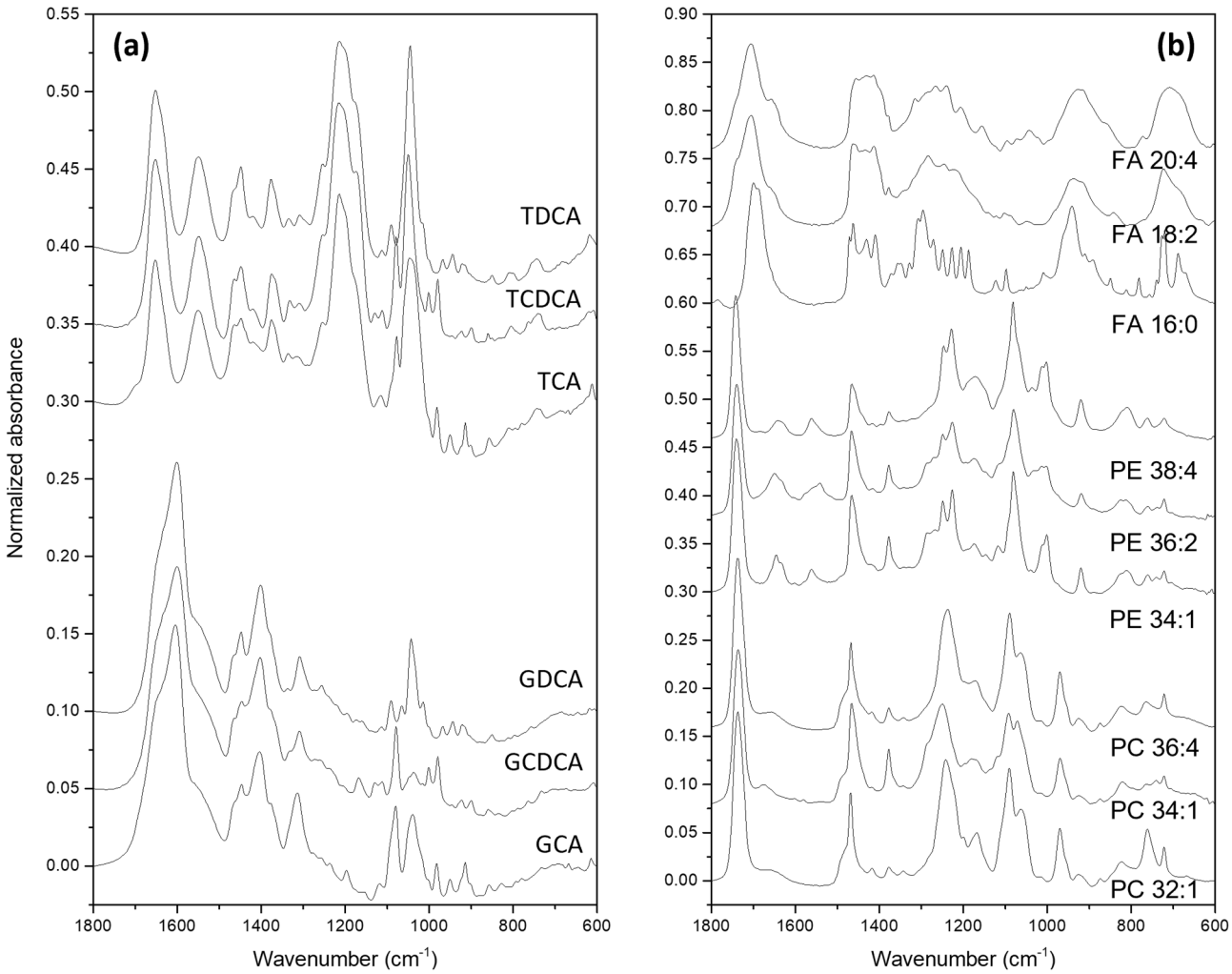

Fig. 2. IR spectra of six bile salts (a), and three PCs, three PEs, and three FAs (b).

TCDCA) in the GB cancer samples were substantially lower (*p*-values smaller than either 0.01, or 0.001) than those in the normal, GB polyp, and HCC samples. The concentration differences in GDCA and TDCA between the normal and GB cancer groups were not significant, but significant between the GB polyp and GB cancer groups (*p*-values smaller than either 0.01 or 0.001). Concentration differences in GDCA and TDCA between the HCC and GB cancer groups remained significant (*p*-values smaller than 0.05), but less distinctive than those of the other four bile salts. Overall, these results indicate that concentrations of bile salts in bile decrease with the development of GB cancer and this decrease in concentration is greater for primary bile salts. A few prior studies have also reported that levels of total and individual bile acids in bile from GB cancer patients were lower than those in patients with GB stones [7,8]. The GB stores and concentrates bile through absorption of water through the epithelial mucosa [9]. The expression of COX-2 (cyclooxygenase-2) in the GB epithelium is elevated in cancer [10,11]; this increase in COX-2 thickens the GB wall and decreases water absorption by the mucosa [12]. This reduced water absorption ability results in dilution of bile, which explains the decreased bile salt concentrations in the case of GB cancer. Meanwhile, it is worthwhile to note that the differences in concentrations of the six bile salts among the other three groups (normal, GB polyp, and HCC) were not significant.

Direct IR measurement of the liquid aqueous phase was not possible because of severe attenuation of IR radiation by strong water absorption. Therefore, we dried aqueous phase samples for IR measurements and an acquired spectrum was normalized to compensate for the thickness variation in the dried sample. Due to the normalization, the lower bile salt content in GB cancer samples may not be fully reflected in the IR spectrum (refer to Fig. 1(a)). To secure the discriminability under this circumstance, the relative abundance of bile salts needs to vary according to the diseases. For examination, the concentrations of bile salts in the normal and GB polyp groups were merged into one group, and the corresponding averages and standard deviations were newly calculated, as shown in Fig. 4(g) (refer to the red squares). Data for the HCC (black) and GB cancer (green) groups are also displayed in this plot.

When the average concentrations of individual bile salts were compared between the merged (normal-GB polyp) and GB cancer groups, the differences were highly significant (*p*-values less than 0.01 or 0.001), as expected. Next, ratios of bile salt concentrations in the merged and GB cancer groups were calculated (Fig. 4(h), green bars). If dilution of bile is only a GB-cancer driven symptom, the concentration ratios of the bile salts should remain constant. However, concentration ratios were different. Concentration ratios of GCA and GCDCA were similarly low (around 5.7), while those of GDCA and TCDCA were high (about 10). Concentration ratios of TCA and TDCA were relatively analogous (approximately 8). Therefore, the relative abundance of individual bile salts differed in the bile samples of GB cancer and was reflected in the corresponding IR spectra (Fig. 1(a) and (b)). These results suggest that bile acid metabolism is likely altered by development of GB cancer.

Meanwhile, there were no significant differences in the concentrations of individual bile salts between the merged and HCC groups (Fig. 4

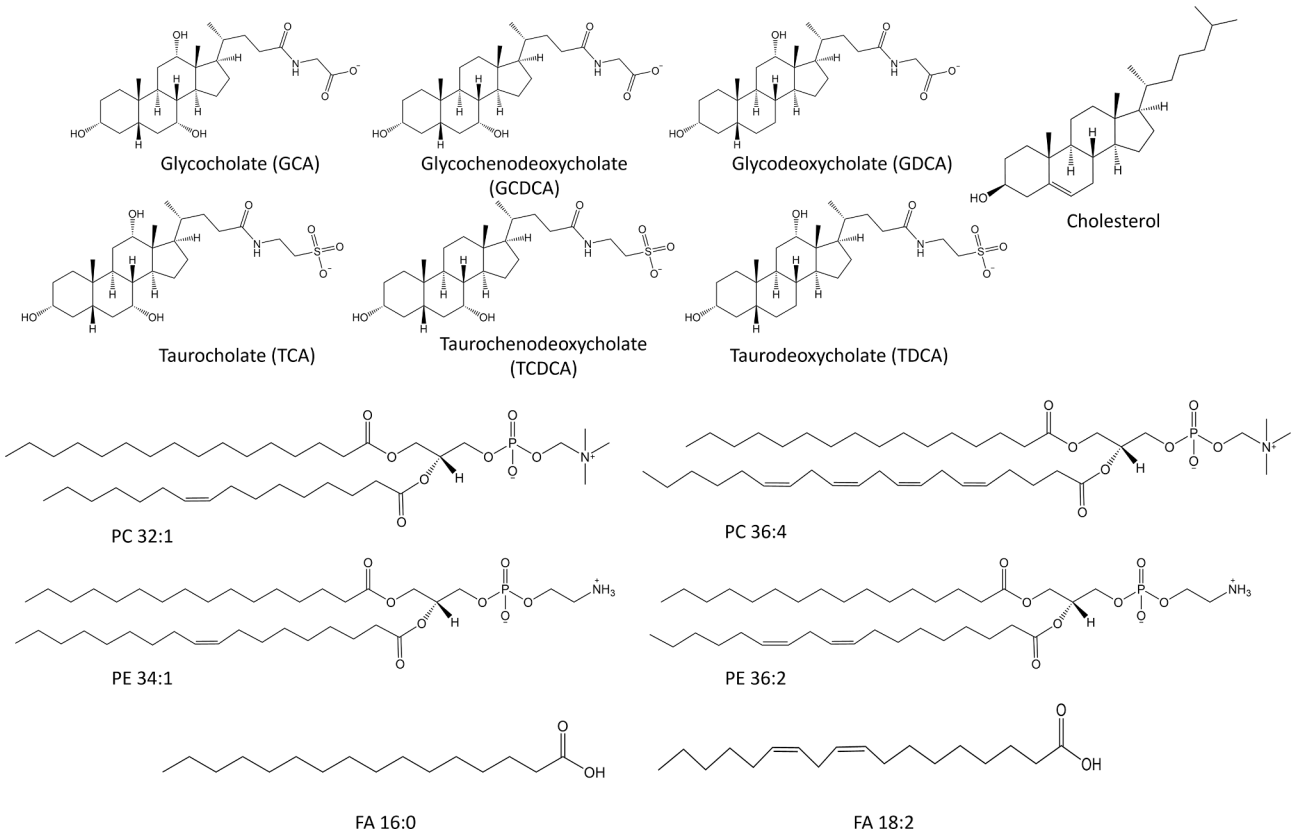

Fig. 3. Molecular structures of selected bile salts, PCs, PEs, FAs, and cholesterol.

(g)). Furthermore, ratios of bile salt concentrations were close to 1 (average ratio: 1.1; black bars in Fig. 4(h)) and their magnitudes did not change significantly. These observations indicate that concentrations of individual bile salts as well as the relative abundance of bile salts in the bile samples of HCC patients were not notably different from those of normal/GB polyp subjects. Therefore, IR analysis of the aqueous bile phase (basically bile salts) cannot identify HCC.

Similar plots to those shown in Fig. 4(g) and (h) were generated for three bile acids (CA, CDCA, and DCA) and two other bile salts (taurohyodeoxycholate (THDCA) and taurolithocholate (TLCA)) (refer to supplementary material). The concentrations of these bile acids and bile salts were considerably lower (below 1.0 mM) than the six abundant bile salts examined in Fig. 4. Differences in the average concentrations of CA, CDCA, and DCA between the merged and GB cancer groups were not significant and their concentration ratios were approximately 0.8. There were significant differences in the concentrations of TLCA and THDCA between the two groups (*p*-values smaller than 0.01) and their concentration ratios ranged from 4 to 7. Nonetheless, the contributions of TLCA and THDCA to the IR spectra of the aqueous phase expect to be minimal due to their low concentrations.

To execute DD-SIMCA, the spectra of normal, GB polyp, and HCC samples were merged together as a training set. Fig. 5(a) shows the acceptance plot resulted using the training set. For DD-SIMCA, the first and second principal components describing most significant spectral variation in the dataset were used with setting β (confidence level) = 0.01. The green curve (boundary) in the plot indicates the acceptance area. As seen, all the training samples are located inside the green curve, indicating that they are within one class without outliers; while, the differentiation of normal, GB polyp, and HCC samples is unrealizable. Fig. 5(b) shows the acceptance plot for the test (GB cancer) samples. Out of the 8 GB cancer samples, 5 samples are non-members (correctly identified as GB cancer, outside of the green curve) and 3 samples are members (incorrectly identified, within the green curve). Although the spectral features of aqueous phases extracted from the bile samples of GB cancer patients were quite different from those of the remained samples, as shown in Fig. 1(a), they were not distinctive enough for the clear discrimination.

3.2. *IR spectral features of organic phases and their discriminability of the diseases*

Fig. 1(c) shows the average IR spectra of the organic phase extracted from the bile samples. The strongest peak at 1738 cm^{-1} corresponds to the C═O group in phospholipids, and the 1465 and 1377 cm^{-1} peaks are from the C—C bonds of cholesterols as well as various lipids [13,14]. Spectra of cholesterol and phosphatidylcholine 32:1 (PC32:1, one of the phospholipids) are also displayed for comparison (refer to the corresponding molecular structures in Figure 3). The mixed shapes of the cholesterol and PC32:1 spectra are analogous to those of the average spectra. Consistent with the analysis findings for the aqueous phase samples, the organic phase spectra of normal, GB polyp, and HCC groups were similar to each other, but dissimilar to that of GB cancer. For example, in the GB cancer spectrum, the 1465 and 1377 cm^{-1} peaks were more intense and band shapes in the 1310–1010 cm^{-1} range (indicated by the horizontal lines) were quite different from those in the spectra of the other groups.

Fig. 1(d) shows the difference spectra of GB polyp, HCC, and GB cancer samples with respect to the normal spectrum. Again, the difference spectra of GB polyp and HCC groups were similar. However, the difference spectrum of GB cancer was dissimilar to those of the other samples and the magnitude of the maximal difference was approximatelyeight times greater than those in the difference spectra of GB polyp and HCC samples. The largest difference was observed around the 1738 cm^{-1} peak where the positive-to-negative difference indicating a shift in peak position occurred. Differences around the 1463, 1377, 1244, and 1092 peaks were also large. This observation indirectly

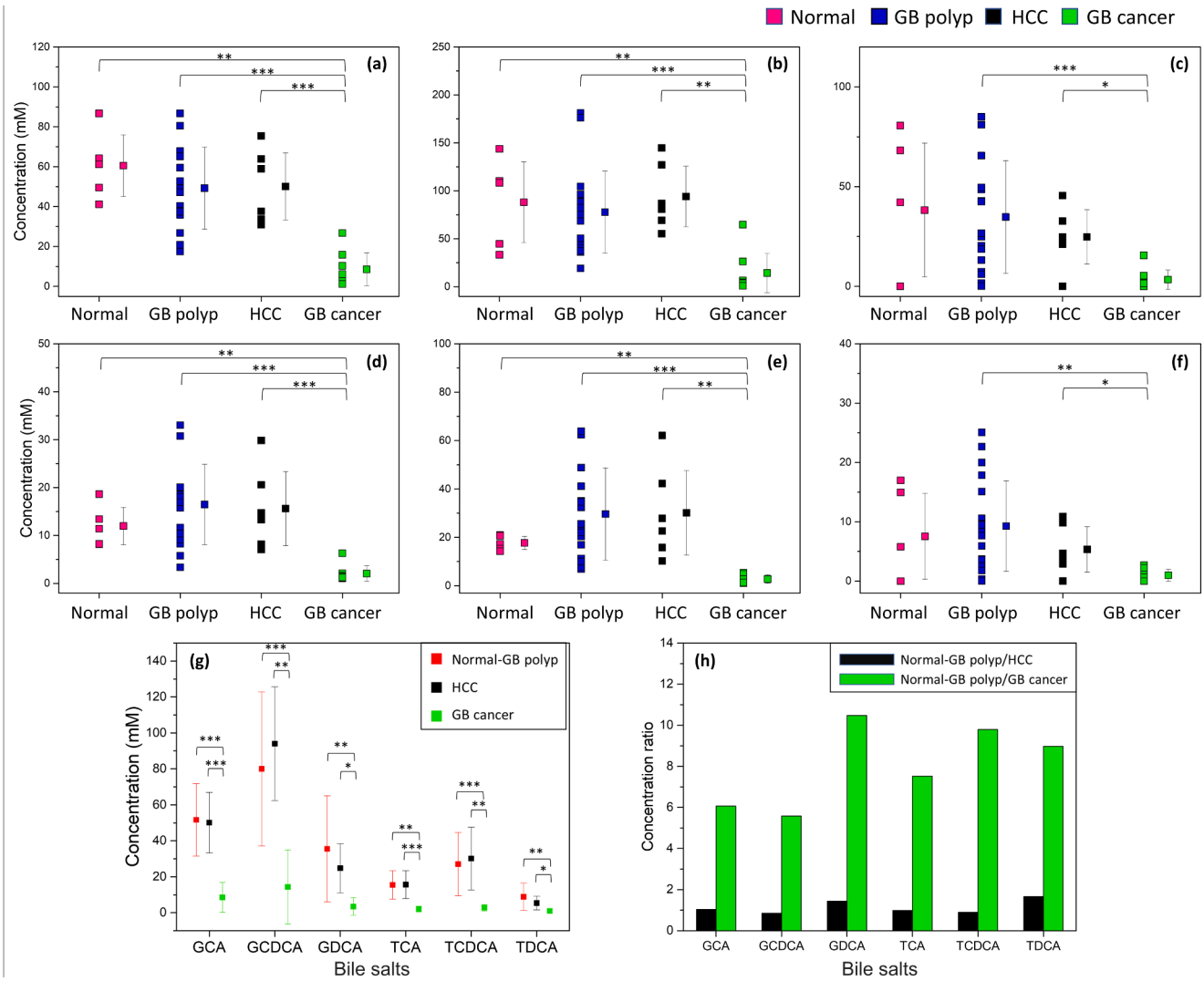

Fig. 4. Concentrations of GCA (a), GCDCA (b), GDCA (c), TCA (d), TCDCA (e), and TDCA (f) in aqueous phase samples of bile collected from normal, GB polyp, HCC, and GB cancer patients. The corresponding averages and standard deviations are also shown. Pairwise group comparisons that yielded *p*-values smaller than 0.05, 0.01, or 0.001 are designated by *, **, or ***, respectively. Averages and standard deviations of the concentrations of each bile salt in the normal-GB polyp merged, HCC, and GB cancer groups (g) and the concentration ratios of each bile salt in the merged/HCC group and merged/GB cancer group (h) are also shown.

indicates that the compositions of hydrophobic constituents in GB cancer samples differ substantially from those in the GB polyp and HCC samples. Therefore, IR spectral features of organic phase samples may be more informative than those of aqueous phase samples for discriminating GB cancer from the other groups.

LC-MS based lipidomic analysis of nine lipid classes (carnitines (Cars), ceramides (Cers), diglycerides (DGs), FAs, lysophosphatidylcholines (LysoPCs), lysophosphatidylethanolamines (LysoPEs), PCs, PEs, and triglycerides (TGs)) in the organic phase samples was performed. One hundred twenty-four lipids (refer to supplementary material) were analyzed in the positive and negative ion modes. According to prior lipidomic studies [15,16], PCs are the most dominant lipids in bile and PEs are the second most abundant. Among the 35 PCs identified, six PCs (PC32:1, PC34:1, PC34:2, PC36:2, PC36:4, and PC38:6) with high peak intensities were selected for further analysis. The IR spectra of PC32:1, PC34:1, and PC36:4 are shown in Fig. 2(b) and the molecular structures of PC32:1 and PC36:4 are provided in Fig. 3.

Fig. 6(a) through (f) show the intensities of the six selected PCs in the organic phase samples of normal, GB polyp, HCC, and GB cancer groups. Corresponding averages and standard deviations, and the results of statistical significance testing are also displayed. Intensities of the six PCs in the GB cancer samples were all significantly lower than those in the normal, GB polyp, and HCC samples. Similar to the results of bile salt analysis, intensity differences of the six PCs were not significantly different among the normal, GB polyp, and HCC groups.

Fig. 6(g) shows the averages and standard deviations of the intensities of each PC in the merged, HCC and GB cancer groups. Differences in the intensities of the six PCs between the two groups were all significant (p-values smaller than 0.05, 0.01 or 0.001). Intensity ratios of each PC in the two groups are presented in Fig. 6(h) (green bar). The intensity ratio of PC36:4 was the highest at 10.2. The intensity ratios of PC34:2, PC36:2, and PC38:6 were relatively similar (range 9.0 to 10.0), while those of PC32:1 and PC34:1 were significantly lower (3.5 and 5.9, respectively). As seen, the intensity ratios of the six PCs were notably different. This indicates that the relative abundance of the six PEs varied considerably between the GB cancer and other samples, and could therefore potentially be used to identify GB cancer. However, there were no significant differences in the intensities of these PCs between the merged and HCC groups. Intensity ratios of each PC between the two groups were very small and their magnitudes were consistent (black bar), as also found in the analysis of bile salts. Therefore, the composition of PCs in bile is not descriptive for identification of HCC.

Among the eight analyzed PEs, six PEs (PE34:1, PE34:2, PE36:2, PE38:4, PE38:6, and PE40:6) were chosen for further investigation. The

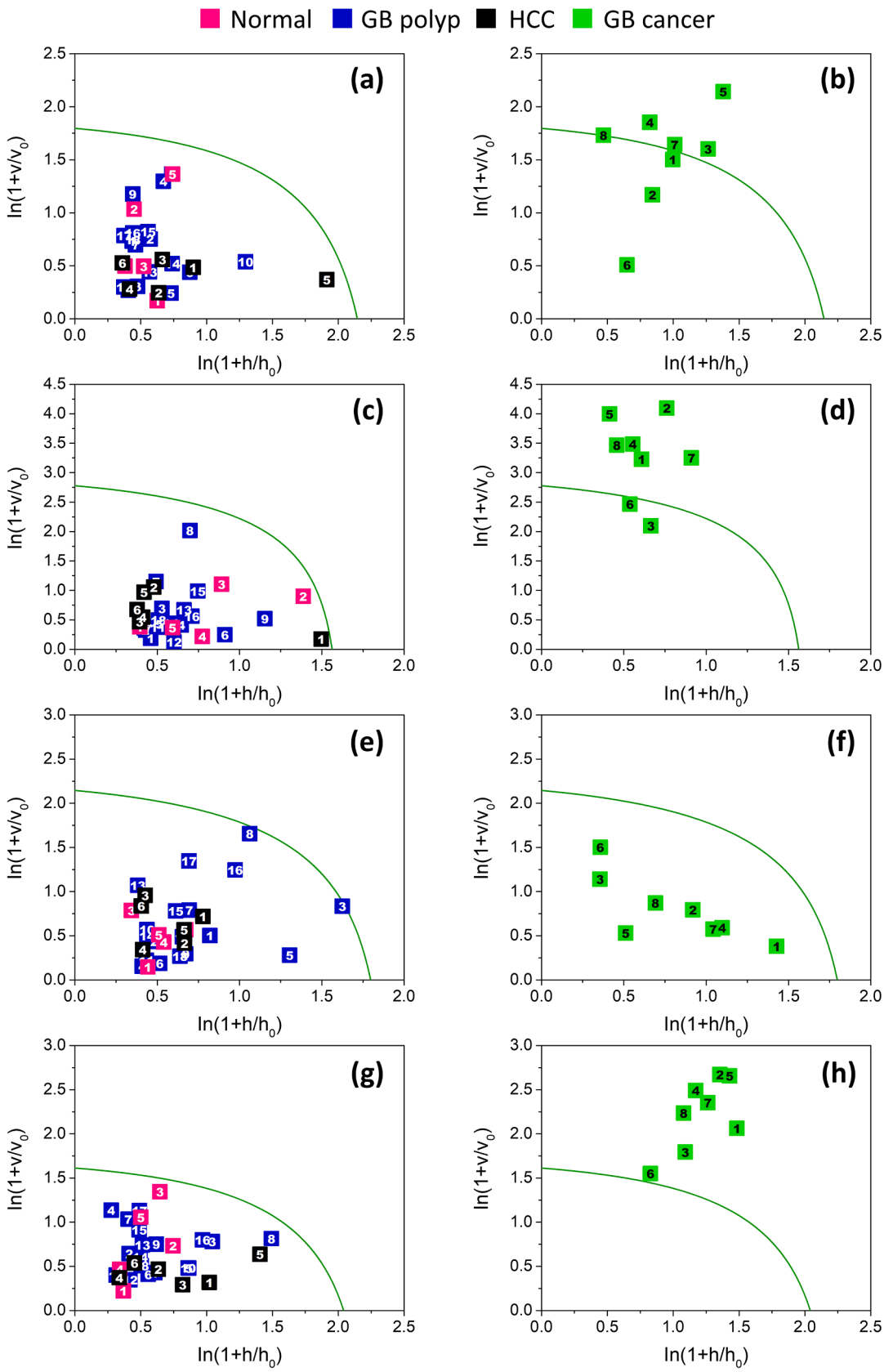

Fig. 5. Acceptance plots for the training and test (GB cancer) samples resulted from using the IR spectra of aqueous ((a), (b)), organic ((c), (d)), and AA ((e), (f)) phase samples. The acceptance plots in the case of employing the three phase-merged spectra ((g), (h)) are also shown.

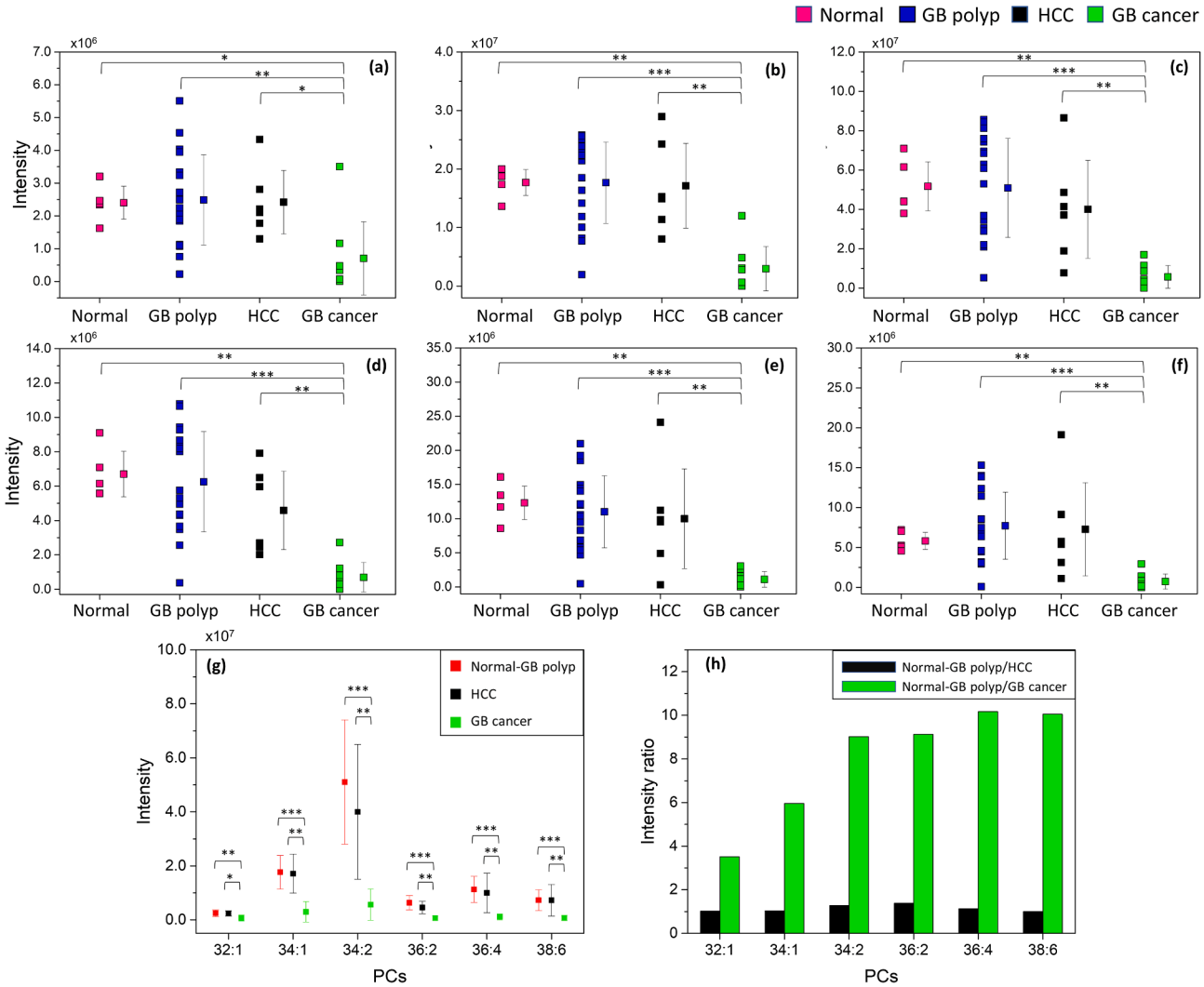

Fig. 6. Intensities of PC32:1 (a), PC34:1 (b), PC34:2 (c), PC36:2 (d), PC36:4 (f), and PC38:6 (g) in organic phase samples of bile collected from normal, GB polyp, HCC, and GB cancer patients. Corresponding averages and standard deviations are also shown. Pairwise comparisons that yielded *p*-values smaller than 0.05, 0.01, or 0.001 are designated by *, **, or ***, respectively. Averages and standard deviations of the intensities of each PC in the normal-GB polyp merged, HCC, and GB cancer groups (g) and comparison of intensity ratios of each PC in the merged/HCC groups and merged/GB cancer groups (h) are also shown.

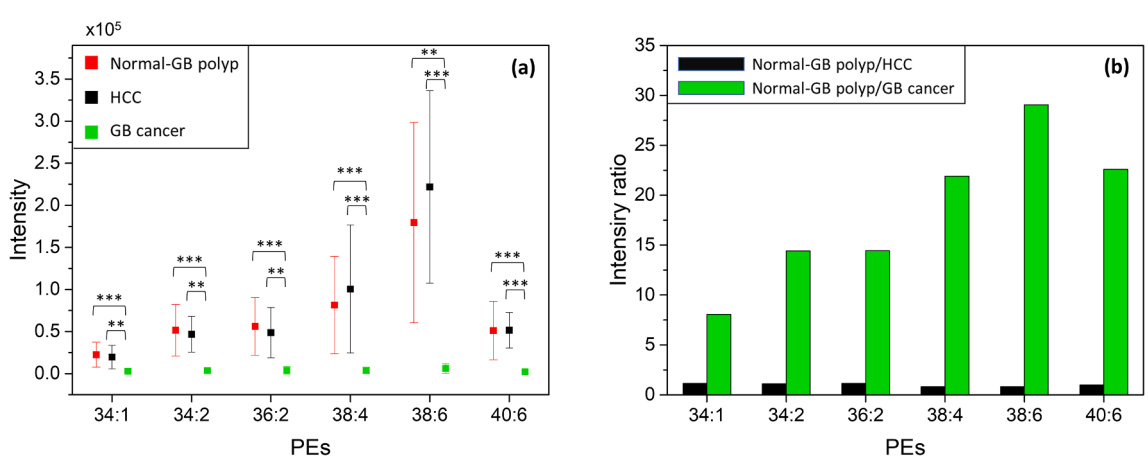

Fig. 7. Averages and standard deviations of the intensities of each PE in the normal-GB polyp merged, HCC, and GB cancer groups (a), and ratios of the average intensity of each PE between the merged-HCC group and between the merged-GB cancer group (b). Pairwise comparisons that yielded *p*-values smaller than 0.05, 0.01, or 0.001 are designated by *, **, or ***, respectively.

IR spectra of PE34:1, PE36:2, and PE38:4 are displayed in Fig. 2(b) and the molecular structures of PE34:1 and PE36:2 are presented in Fig. 3. Fig. 7(a) shows the averages and standard deviations of the intensities of each PE in the merged, HCC and GB cancer groups. Differences in the intensities of the six PEs between the two groups were all significant (*p*-values smaller than either 0.01 or 0.001). Fig. 7(b) shows the intensity ratios of PEs in the two groups. Intensity ratios of the six PEs varied widely. Notably, the corresponding average ratio was 18.4, which is much greater than the average of 8.0 in the analysis of PCs (Fig. 6(h)). Substantial decreases in PE concentrations and large changes in the relative abundance of PEs in bile may allow IR-based discrimination of GB cancer. Again, the intensity differences in each PE between the merged and HCC groups were not significant and the corresponding intensity ratios did not vary noticeably.

The intensities of six different FAs (FA16:0, FA18:0, FA18:1, FA18:2, FA20:4, and FA22:6) were also investigated. The IR spectra of FA16:0, FA18:2, and FA20:4 are shown in Fig. 2(b) and the molecular structures of FA16:0 and FA18:2 are displayed in Fig. 3. Fig. 8(a) shows the averages and standard deviations of the intensities of each FA in the merged, HCC and GB cancer groups. Interestingly, in contrast to what was observed for PCs and PEs, the average peak intensities of the six FAs in the GB cancer samples were slightly higher than those of the merged samples; while, the intensity differences were only significant for FA18:1, FA20:4, and FA22:6 (*p*-values smaller than 0.05). Subsequently, the variation among the intensity ratios of each FA was relatively small (Fig. 8(b)). Intensities of individual FAs were not significantly different between the merged and HCC groups.

Plots similar to those shown in Fig. 8(a) and (b) were examined for the remaining lipid classes (Cars, Cers, DGs, LysoPCs, LysoPEs, and TGs) by selecting 2–6 main components in each lipid class (see supplementary material). Differences in the intensities of Cars and Cers between merged and GB cancer groups were significant and the corresponding intensity ratios were dissimilar. For the remaining four lipid classes, differences in the corresponding intensities between the two groups were not significant.

Fig. 5(c) and (d) presents the acceptance plots for the training and test samples, respectively, when the IR spectra of organic phase samples are used for DD-SIMCA. Again, the first and second principal components were used with setting $\beta = 0.01$. All the training samples are again located inside the green curve without outliers. In the case of test samples (Fig. 5(d)), six and two GB cancer samples were correctly (non-members) and incorrectly (members) identified, respectively. Although the spectral features of GB cancer samples were considerably different from those of remained samples due to the dissimilar lipid compositions, specifically PCs, PEs, Cars, and Cers, the differences were again not sufficient for the complete isolation of GB cancer samples.

3.3. IR spectral features of AA phases and their discriminability of the diseases

Fig. 1(e) shows the average IR spectra of AA phases from the bile samples. Amide I and II bands corresponding to protein secondary structures appeared at 1653 and 1543 cm^{-1}, respectively; other side chain bands were present in the 1450–1200 cm^{-1} range [17]. Overall, the spectra of the four groups were similar, but the intensity of the 1071 cm^{-1} peak corresponding to PO^{2-} vibration of the phosphate group of phosphorylated protein molecule [18] was less intense in the GB cancer spectrum than spectra from the other samples. Difference spectra of GB polyp, HCC, and GB cancer samples with respect to the normal spectrum are shown in Fig. 1(f). Again, the GB polyp and HCC difference spectra were similar to each other but dissimilar to the difference spectrum of GB cancer. More positive difference peaks were present in the GB cancer spectrum than the other spectra. This confirmed that GB cancer samples had dissimilar AA spectral features to the other samples, suggesting that spectral features could be useful for discrimination of GB cancer samples from the other samples.

Fig. 5(e) and (f) display the acceptance plots for the training and test samples, respectively, when the IR spectra of AA phase samples are employed for DD-SIMCA. Also, the first and second principal components were used with setting $\beta = 0.01$. All the training samples are again located inside the green curve without outliers, while all the 8 GB cancer samples were incorrectly classified as members. Although the AA phase spectrum of GB cancer was somewhat characteristic, the dissimilarity was too small to differentiate the GB cancer samples from the other samples.

In our previous work, we showed that discrimination between GB polyp and GB cancer samples in the principal component score domain improved when IR spectra obtained from the three separated phases were merged compared to the use of IR spectra of each separated phase individually [1]. Therefore, we performed DD-SIMCA using the merged spectra from the three phases as shown in Fig. 5(g) and (h). The first four principal components were used with setting $\beta = 0.01$. All the training samples are located inside the green curve without outliers and all the 8 GB cancers samples were correctly classified as non-members. Since the IR spectral features of each separated phase were mutually characteristic and complementary to represent different metabolites in bile, the use of three phase-merged spectra was again cooperative for the clear discrimination of GB cancer.

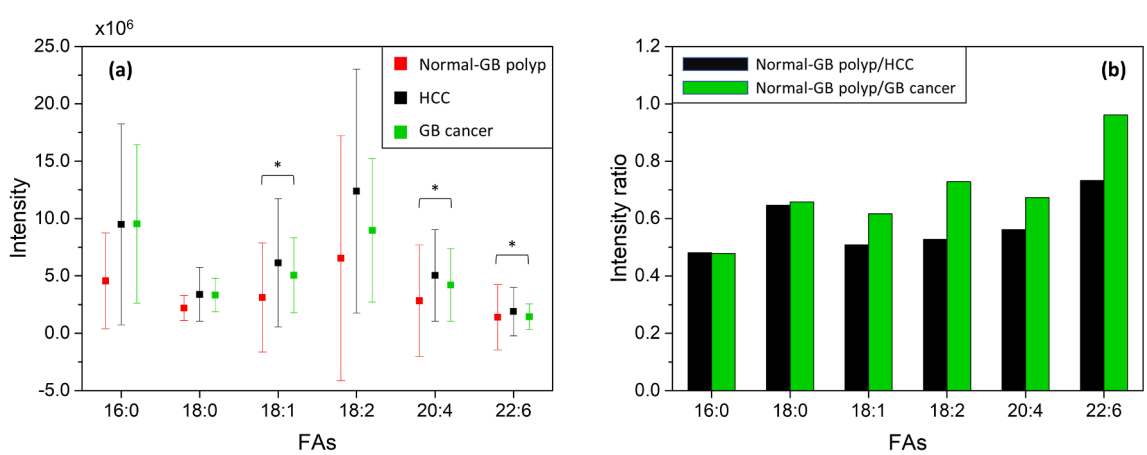

Fig. 8. Averages and standard deviations of the intensities of each FA in the normal-GB polyp merged, HCC, and GB cancer groups (a), and ratios of the average intensity of each FA between the merged-HCC group and merged-GB cancer group (b). Pairwise comparisons that yielded *p*-values smaller than 0.05, 0.01, or 0.001 are designated by *, **, or ***, respectively.

4. Conclusion

Based on the analysis of IR spectra obtained from three different extracted phases of 37 diverse bile samples, the spectral features of each separated phase were distinctly characteristic and the use of three phase-merged spectra for DD-SIMCA enabled to clearly discriminate GB cancer samples from normal, GB polyp and HCC samples. The discrimination between HCC samples and normal/GB polyp samples was not realized due to the similar bile composition of the corresponding samples. Unlike GB cancer, HCC does not appear to significantly alter the composition of bile constituents. Two issues warrant future study. First, more extensive analysis of other major components of bile such as cholesterols and proteins is necessary to further explain the origin for the dissimilar spectral feature of GB cancer sample. Second, a larger sample set with a balanced distribution of sample groups is beneficial to assess the further clinical utility of the demonstrated IR analysis scheme. We are currently collecting samples, but progress is slow as bile is not a frequently obtained specimen unlike blood. When IR information from the separated phases of a large sample set is ultimately available, diverse deep learning methods combined with feature extraction strategies will be beneficial for accurate identification of GB cancer.

CRediT authorship contribution statement

Eunjin Jang: Investigation, Methodology, Software, Data curation, Formal analysis, Visualization, Writing – review & editing. **Sunhee Jung:** Investigation, Data curation, Methodology. **Woosuk Sohng:** Formal analysis, Software, Validation. **Dongho Choi:** Conceptualization, Resources. **Geum-Sook Hwang:** Conceptualization, Writing – original draft. **Hoeil Chung:** Project administration, Funding acquisition, Conceptualization, Writing – original draft, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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